

CPU vs GPU Comparison in AI Model Inference

Introduction

In artificial intelligence, model execution can be performed using either a Central Processing Unit (CPU) or a Graphics Processing Unit (GPU). Both play an important role in running AI models, especially during the inference phase. This comparison helps in understanding their performance, efficiency, and suitability in different scenarios.

CPU vs GPU Comparison

A CPU is a general-purpose processor designed to handle a wide variety of tasks sequentially. It executes instructions one after another, making it suitable for basic operations and environments where computational demand is low. In contrast, a GPU is specifically designed for parallel processing and can perform multiple computations simultaneously. This makes it highly efficient for handling large-scale data and complex AI models.

During the implementation of this project, AI models were tested in both CPU and GPU environments using Google Colab. It was observed that CPU-based inference was slower, typically taking around 10 to 30 seconds depending on the model and input size. On the other hand, GPU-based inference was significantly faster, completing the same tasks in approximately 3 to 8 seconds. This difference is mainly due to the GPU's ability to process multiple operations in parallel.

Despite the difference in speed, the output generated by the model remained consistent in both CPU and GPU environments. This indicates that while GPUs improve performance, they do not affect the accuracy or correctness of the results.

Advantages of GPU

The GPU offers several advantages, particularly in AI applications. It provides faster execution due to parallel processing capabilities and is highly suitable for large models and heavy computations. GPUs significantly reduce inference time, making them ideal for real-time applications and large-scale deployments.

Advantages of CPU

The CPU, although slower, is more widely available and does not require specialized hardware. It is useful in scenarios where GPU resources are not available or when working with smaller models. CPU-based inference is also more cost-effective and can still produce correct results, making it suitable for basic applications and testing purposes.

Conclusion

In conclusion, both CPU and GPU have their own advantages in AI model inference. GPUs provide faster performance and are preferred for handling complex and large-scale models, while CPUs offer accessibility and cost-effectiveness. The choice between CPU and GPU depends on the specific requirements of the application, such as performance needs, resource availability, and budget constraints.